

Achintya Thokur Ravi

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Academic

B.Tech in Computer Science & Engineering (Artificial Intelligence) 2024–2028
(2nd Year Completed)
Amrita Vishwa Vidyapeetam, Chennai CGPA: 8.29

Skills

Languages: Python, Java, C
ML/AI: Gaussian Process Regression, Graph Neural Networks (GCN, GIN)
Libraries: PyTorch, Scikit-Learn, NumPy, Pandas
Tools: Git, FastAPI
Domains: Scientific Machine Learning, Probabilistic Modeling, Computational Biology

Projects

Kronecker-Structured Gaussian Processes for Scalable Regression 2026
github.com/AchintyaTR/KroncekerStructuredGP

- Implemented Gaussian Process regression using Kronecker-structured covariance matrices to exploit separable kernel structure in high-dimensional inputs.
- Reduced computational bottlenecks of standard GP inference ($\mathcal{O}(n^3)$) by leveraging tensor algebra and structured matrix operations.
- Developed numerically stable inference routines using structured linear algebra (e.g., eigendecomposition / Cholesky factorization) for scalable regression.
- Enabled efficient modeling of grid-like and spatiotemporal data through covariance decomposition, improving memory and computational efficiency.

Protein Thermal Stability Prediction using Sequence & Structural Features 2026
github.com/AchintyaTR/TemStabilityProtein

- Built a machine learning pipeline to predict protein melting temperature (T_m) using sequence embeddings and structural features.
- Integrated protein language model representations with regression models to capture sequence-level dependencies.
- Incorporated structural information via engineered features (e.g., residue-level or graph-based representations) to model 3D spatial relationships.
- Evaluated model performance using regression metrics such as RMSE and correlation, following standard practices in computational protein modeling.

Research Experience

Researcher — Machine Learning & Autonomous Systems 2026–Present
Amrita Vishwa Vidyapeetam

- Implementing probabilistic state estimation methods including Kalman Filters and nonlinear variants for dynamic systems.
- Exploring structured machine learning approaches for robotics and time-varying systems.

Achievements

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- Strong academic performance in AI and Machine Learning coursework.
 - Built research-oriented ML systems bridging theoretical methods with real-world applications.